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## DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
## MINI PROJECT (BCS586) – REVIEW 0

# NETSHIELD

## Machine Learning-Based Intrusion Detection System

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# INTRODUCTION

Networks today carry more traffic, more devices and more threats than ever. For our Mini Project (BCS586) in the Networking domain, we studied three ideas, each backed by a recent IEEE paper, compared them, and selected one.

1

### 6G Network Digital Twin

AI-driven digital twin that predicts traffic and self-optimises 6G networks.

2

### INTLOWEND

Programmable low-end networking using P4 over LoRa/LPWAN links.

3

### NetShield

Machine learning-based intrusion detection system for network traffic.

**TODAY'S FLOW**

Ideas | Comparison | Why NetShield | What's Different | Conclusion

# IDEA 1: 6G DIGITAL TWIN

*AI-Native Network Digital Twin for Self-Optimizing 6G Networks*

## PROBLEM STATEMENT

6G networks must handle highly dynamic traffic, ultra-low latency and better energy efficiency. Traditional management struggles with these changing conditions.

## PROPOSED APPROACH

- Virtual replica of the mobile network
- Traffic prediction & anomaly detection
- MLP-based latency prediction
- Adaptive load & energy optimisation

## TOOLS & TECHNOLOGY

Python, scikit-learn, Pandas, NumPy, Matplotlib. Isolation Forest and MLP models on the Telecom Italia Milan Grid dataset.

## LIMITATIONS

- Limited time period and locations
- Derived KPIs may differ from live networks
- Large-scale real-time use not evaluated

## REFERENCE PAPER

J. Sengendo and F. Granelli, “Toward Self-Optimizing 6G Networks Through Network Digital Twin Intelligence: A Real Network Traffic Evaluation,” IEEE Networking Letters, vol. 8, 2026.

**98.3%**
traffic prediction accuracy (voice/SMS)

# IDEA 2: INTLOWEND

*INTLOWEND: Programmable Low-End Networking using P4 and LPWAN*

## PROBLEM STATEMENT

LPWAN devices are vendor-locked with fixed packet processing. The goal is a low-cost, low-power, long-range system whose packet parsing can be redefined in software.

## PROPOSED APPROACH

- 3 layers: Service, Programmable, Link
- P4 for packet parsing & match-action
- LoRa link for long-range communication
- Protocol simplification + compression

## TOOLS & TECHNOLOGY

P4/P4c, BMv2, Raspberry Pi 4, LoRa GPS HAT (SX127x), PiSugar battery. DHCP, DNS and NAT via udhcpd, bind9, iptables.

## LIMITATIONS

- Half-duplex LoRa; sync challenges
- 255-byte MTU needs fragmentation
- Short-range, controlled testing only
- Raspberry Pi not final low-power HW

## REFERENCE PAPER

V. B. Alves et al., “Towards Programmable Low-End Networking: Research Challenges and Lessons Learned,” IEEE/IFIP NOMS 2025.

**3 layers**

Service · Programmable · Link

# IDEA 3: NETSHIELD

SELECTED

*NetShield: Machine Learning-Based Intrusion Detection System*

## PROBLEM STATEMENT

Signature-based IDS miss new and evolving threats. We need an automated ML system that flags abnormal traffic, cuts false alarms and supports real-time monitoring.

## PROPOSED APPROACH

- Preprocessing and feature selection
- Train and compare 5 ML classifiers
- Classify traffic as normal or anomalous
- Accuracy, precision, recall, F1-score

## TOOLS & TECHNOLOGY

Python, scikit-learn, Pandas, NumPy, Matplotlib/Seaborn. Flask/FastAPI backend, React.js dashboard, SQLite/PostgreSQL.

## LIMITATIONS

- Depends on dataset quality & diversity
- Offline tests differ from real networks
- Unseen attacks and false alarms possible
- Live packet capture not yet implemented

## REFERENCE PAPER

M. Prasanna Venkatesh et al., “AI Powered Network Intrusion Detection System using Machine Learning,” ICCCES-2026, Chennai, pp. 195–200.

**5 ML models**

LR · AdaBoost · RF · XGBoost · SVM

# COMPARISON OF THE THREE IDEAS

| Criteria | 6G Digital Twin | INTLOWEND | NetShield |
|---|---|---|---|
| **Focus** | 6G network management (AI) | Programmable low-end networking | Network security (ML-based IDS) |
| **Core technique** | Isolation Forest, MLP, digital twin | P4, BMv2, LoRa/LPWAN | LR, AdaBoost, RF, XGBoost, SVM |
| **Hardware needed** | Laptop, 8 GB RAM | Raspberry Pi 4, LoRa HAT, battery, power meter | Laptop, 8 GB RAM, 10 GB storage |
| **Data / setup** | Telecom Italia Milan Grid dataset | Physical testbed and measuring tools | Public labelled traffic datasets |
| **Evaluation** | Traffic, latency, load, energy KPIs | Latency, goodput, power use | Accuracy, precision, recall, F1 |
| **Main challenge** | Limited data; real-time unproven | Half-duplex link, 255-byte MTU | Unseen attacks, false alarms |
| **Mini-project fit** | **Medium** | **Low to medium** | **High** |

*Mini-project fit is our assessment, based on hardware, data and time needed.*

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# WHY WE CHOSE NETSHIELD

### Real-world relevance
Signature-based IDS miss new threats; ML can learn to flag them.

### Low cost, laptop-ready
Runs on a laptop with open-source tools; INTLOWEND needs a LoRa/Pi testbed.

### Open, public datasets
Public labelled datasets exist; the 6G twin relies on one limited-period telecom dataset.

### Measurable results
Five models compared using accuracy, precision, recall, F1 and confusion matrix.

### Demo-ready dashboard
A web dashboard shows detection results and network activity.

### Easy to extend
Modular design: add models, live packet capture and real-time monitoring later.

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# WHAT'S DIFFERENT IN NETSHIELD

| Existing solution (reference paper) | NetShield (our project) |
| --- | --- |
| Offline model training and evaluation on a dataset | Adds a dashboard to visualise detection results and network activity |
| Results reported as metrics and model comparisons | Modular system: Flask/FastAPI backend, React dashboard, SQL database and auto-generated reports |
| Live packet capture left as future work | Built for extension to live packet capture and real-time monitoring (planned) |
| Limited attack types and feature engineering | Extensible with new models, attack types and improved features |

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# CONCLUSION

## KEY TAKEAWAYS

1. We studied three IEEE-backed ideas: 6G Digital Twin, INTLOWEND and NetShield.
2. NetShield is the most feasible: laptop-ready, open-source, with clear metrics.
3. It goes beyond the reference paper with a dashboard, modular backend and a path to live monitoring.

## NEXT STEPS

1. Collect and preprocess a labelled traffic dataset
2. Train and compare the five ML models
3. Build the backend API and dashboard
4. Evaluate, report and extend to live capture

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# Thank You

Your Questions, Ideas & Suggestions are Welcome



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